

Quantifier-Free Linear Arithmetic

Because my mathematics has its origin in a real problem doesn't make it less interesting to me — just the other way around, I find it makes the puzzle I am working on all the more exciting. I get satisfaction out of knowing that I'm working on a relevant problem.

— George Dantzig

*An Interview with George B. Dantzig:
The Father of Linear Programming, 1986*

This chapter considers satisfiability in the quantifier-free fragment of the theory of rationals $T_{\mathbb{Q}}$. Addressing this fragment is motivated by two observations. First, program verification typically requires just considering formulae from the quantifier-free fragments of theories such as $T_{\mathbb{Q}}$. Second, deciding satisfiability in the full theory of $T_{\mathbb{Q}}$ is computationally expensive, while deciding satisfiability in just the quantifier-free fragment of $T_{\mathbb{Q}}$ is fast in practice when using, for example, the **simplex method** for **linear programming**.

A linear program is an optimization problem in which the goal is to find a point satisfying a set of linear *constraints* that maximizes a linear *objective function*. The linear constraints are a quantifier-free conjunctive $\Sigma_{\mathbb{Q}}$ -formula in which each atom is a weak inequality. The objective function is a linear function. Deciding $T_{\mathbb{Q}}$ -satisfiability can be cast as solving a linear program, so we benefit from existing algorithms for solving them.

Section 8.1 motivates studying the quantifier-free fragments of theories in general. Section 8.2 reviews concepts from linear algebra. Then Section 8.3 introduces linear programs and shows how to decide $T_{\mathbb{Q}}$ -satisfiability by solving a linear program. Finally, Section 8.4 presents the simplex method for solving linear programs.

8.1 Decision Procedures for Quantifier-Free Fragments

The time complexities of the algorithms of Chapter 7 limit their practical impact. Additionally, Ferrante and Rackoff's method is not optimal for the

quantifier-free fragment of $T_{\mathbb{Q}}$. For other theories, the situation is worse: satisfiability in the full theories such as equality T_E , lists T_{cons} , and arrays T_A is undecidable. Fortunately, for verification and other applications, we typically need to decide satisfiability of quantifier-free formulae rather than of arbitrarily quantified formulae.

Recall that the **quantifier-free fragment** of a theory T with signature Σ consists of the axioms of T and valid Σ -formulae of the form

$$G : \forall x_1, \dots, x_n. F[x_1, \dots, x_n] ,$$

where F is quantifier-free and $\text{free}(F) = \{x_1, \dots, x_n\}$. While such formulae have quantifiers, the point is that they do not have **quantifier alternations**: all quantifiers are universal. Using our conventions, we would ordinarily ask whether the formula F is T -valid — whether its universal closure $\forall * . F$ is T -valid. F is indeed quantifier-free.

T -validity of G corresponds to T -unsatisfiability of

$$\neg G : \exists x_1, \dots, x_n. \neg F[x_1, \dots, x_n] ,$$

or, using our conventions, simply T -unsatisfiability of $\neg F$. Hence, the quantifiers are “natural” for satisfiability checking: $\neg G$ is T -satisfiable iff there exists a T -interpretation I and an assignment to $x_1, \dots, x_n$ under which $\neg F$ evaluates to true.

Fortunately, the quantifier-free fragments of many theories are decidable, often efficiently so. In Chapters 8-10, we focus on decision procedures for the quantifier-free fragments of theories.

For ease of exposition, we consider only **conjunctive** quantifier-free Σ -formulae in each theory T that we examine. Conjunctive Σ -formulae are conjunctions of Σ -literals. This restriction does not limit the scope of the decision procedures. For given arbitrary quantifier-free Σ -formula F , we can convert it into DNF Σ -formula

$$F_1 \vee \dots \vee F_k$$

in which each F_i is conjunctive. F is T -satisfiable iff at least one F_i is T -satisfiable. Decide the T -satisfiability of F by considering each F_i .

Remark 8.1 (★Complexity). This restriction does, however, affect complexity. Because satisfiability in PL is NP-complete, any decision procedure that considers arbitrary quantifier-free formulae must be at least NP-hard, as it must handle not only the theory-specific aspects of the formulae but also the combinatorial (PL) aspects. However, considering only conjunctive formulae allows us to give more insightful complexity bounds. For example, satisfiability in the conjunctive quantifier-free fragments of $T_{\mathbb{Q}}$, T_E , T_{cons} , and their union theory is in PTIME. Thus, the “hard” part of deciding satisfiability of arbitrary quantifier-free formulae in these theories is handling the underlying PL structure. Analyzing the theory-specific aspects is comparatively easy.

8.2 Preliminary Concepts and Notation

We define basic concepts and notation of linear algebra, covering only what is required for understanding the remainder of the chapter. We refer the reader interested in learning more about linear algebra to relevant texts in **Bibliographic Remarks**.

Basic Concepts and Notation

A **variable n -vector** $\bar{x}$ is a column of n variables $x_1, \dots, x_n$. An **n -vector** is a column $\bar{a} \in \mathbb{Q}^n$ of n rationals, and its **transpose** $\bar{a}^T$ is a row with elements listed in the same order:

$$\bar{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \quad \text{and} \quad \bar{a}^T = [a_1 \cdots a_n] \quad .$$

An $m \times n$ -**matrix** $A \in \mathbb{Q}^{m \times n}$ consists of n columns of m rationals each (alternatively, m rows of n rationals each), and its **transpose** A^T is an $n \times m$ -matrix in which element a_{ij} is swapped with element a_{ji} :

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad A^T = \begin{bmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & & \vdots \\ a_{1n} & \cdots & a_{mn} \end{bmatrix} \quad .$$

When we refer to a row $\bar{a}_i$ of A , we mean the row vector $[a_{i1} \cdots a_{in}]$, and when we refer to a column $\bar{a}_j$ of A , we mean the column vector $[a_{1j} \cdots a_{mj}]^T$. We use this compact notation of transposed row vectors for column vectors to save vertical space.

Vector-vector multiplication works as follows:

$$\bar{a}^T \bar{b} = [a_1 \cdots a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^n a_i b_i \quad .$$

Matrix-vector multiplication works as follows:

$$A\bar{x} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n a_{1i} x_i \\ \vdots \\ \sum_{i=1}^n a_{mi} x_i \end{bmatrix} \quad .$$

Each row of A multiplies (as a vector) $\bar{x}$. Finally **matrix-matrix multiplication** works as follows: the product P of AB , for $m \times n$ -matrix A and $n \times \ell$ -matrix B , is an $m \times \ell$ -matrix in which element p_{ij} is the product of the vector-vector multiplication of row $\bar{a}_i$ of A and column $\bar{b}_j$ of B :

$$p_{ij} = \bar{a}_i \bar{b}_j = [a_{i1} \cdots a_{in}] \begin{bmatrix} b_{1j} \\ \vdots \\ b_{nj} \end{bmatrix} = \sum_{k=1}^n a_{ik} b_{kj} .$$

There are several important named vectors and matrices. $\bar{0}$ is a (column) vector of 0s. Similarly, $\bar{1}$ is a vector of 1s. Their sizes depend on the context in which they are used. Combining our notation so far,

$$\bar{1}^\top \bar{x} = \sum_{i=1}^n x_i .$$

In this context, $\bar{1}$ is an n -vector. The $n \times n$ -matrix I is the **identity matrix**, in which the diagonal elements are 1 and all other elements are 0. Thus,

$$IA = AI = A$$

for any $n \times n$ -matrix A . Finally, the **unit vector** $\mathbf{e}_i$ is the vector in which the i th element is 1 and all other elements are 0. Again, the sizes of I and $\mathbf{e}_i$ depend on their context.

Linear Equations

A **vector space** is a set of vectors that is closed under addition and scaling of vectors: if $\bar{v}_1, \dots, \bar{v}_k \in S$ are vectors in vector space S , then also

$$\lambda_1 \bar{v}_1 + \cdots + \lambda_k \bar{v}_k \in S$$

for $\lambda_1, \dots, \lambda_n \in \mathbb{Q}$. $\bar{0}$ is always a member of a vector space. For example, $\mathbb{Q}^n$ is a vector space with **dimension** n ; it consists of all n -vectors of rationals. In general, the dimension of a vector space is given by the minimal number of vectors required to produce all vectors of the space through addition and scaling. Such a minimal set is called a **basis**. For example, a line has dimension 1 and can be described by a single vector: the full space of the line is described simply by scaling this vector. Similarly, a plane has dimension 2 and can be described by two vectors. The vector space consisting of just $\bar{0}$ has dimension 0 and is described by the empty set of vectors.

The **linear equation**

$$F: A\bar{x} = \bar{b} ,$$

for $m \times n$ -matrix A , variable n -vector $\bar{x}$, and m -vector $\bar{b}$, compactly represents the $\Sigma_{\mathbb{Q}}$ -formula

$$F : \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n = b_i .$$

The satisfying points comprise a vector space.

To solve linear equations (at least on paper), we apply **Gaussian elimination**. Consider first the case when A is a **square matrix**: its numbers of columns and rows are equal. For equation $A\vec{x} = \vec{b}$ define the **augmented matrix** $[A \mid \vec{b}]$. A matrix is in **triangular form** if all of its entries below the diagonal are 0; an augmented matrix $[A \mid \vec{b}]$ is in triangular form if A is in triangular form. The goal of Gaussian elimination is to manipulate $[A \mid \vec{b}]$ into triangular form via the following **elementary row operations**:

1. Swap two rows.
2. Multiply a row by a nonzero scalar.
3. Add one row to another.

The last two operations are often combined to yield the composite operation: add the scaling of one or more rows (by nonzero scalars) to a row.

Once an augmented matrix is in triangular form, solving the equations is simple. Solve the equation of the final row, which involves only one variable. Then substitute the solution into the other rows, yielding another equation with one variable. Continue until a solution is obtained for each variable.

Example 8.2. Solve

$$\begin{bmatrix} 3 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 2 \end{bmatrix} .$$

Construct the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 6 \\ 1 & 0 & 1 & 1 \\ 2 & 2 & 1 & 2 \end{array} \right] .$$

Apply the row operations as follows:

1. Add -2 times the first row and 4 times the second row to the third row:

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 6 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -6 \end{array} \right]$$

2. Add -1 times the first row and 2 times the second row to the second row:

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 6 \\ 0 & -1 & 1 & -3 \\ 0 & 0 & 1 & -6 \end{array} \right]$$

This augmented matrix is in triangular form.

Now solve the final row, representing equation $x_3 = -6$, for x_3 , yielding $x_3 = -6$. Substituting into the second equation yields $-x_2 - 6 = -3$, or $x_2 = -3$. Substituting the solutions for x_2 and x_3 into the first equation yields $3x_1 - 3 - 12 = 6$, or $x_1 = 7$. Hence, the solution is $\bar{x} = [7 \ -3 \ -6]^T$. ■

Gaussian elimination can also be applied when A is not a square matrix. Rather than achieving a triangular form, the goal is to achieve **echelon form**, in which the first nonzero element of each row is to the right of those above it. In this case, there are multiple solutions to the equation.

Example 8.3. Suppose that an equation over variables x_1, x_2, x_3, x_4 reduces to the following echelon form:

$$\left[\begin{array}{cccc|c} 3 & 1 & 2 & 0 & 6 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 2 & -6 \end{array} \right]$$

From the last row, $x_4 = -3$. We cannot solve for x_3 because there is not a row in which the x_3 column has the first non-zero element; therefore, x_3 can take on any value. To solve the second row, $-x_2 + x_3 - x_4 = 0$, for x_2 , replace x_4 with its value -3 and let x_3 be any value: $-x_2 + x_3 + 3 = 0$. Then $x_2 = 3 + x_3$. Substituting for x_2 in the first equation, solve $3x_1 + (3 + x_3) + 2x_3 = 6$ for x_1 : $x_1 = 1 - x_3$. Solutions thus lie on the line described by

$$\begin{bmatrix} 1 - x_3 \\ 3 + x_3 \\ x_3 \\ -3 \end{bmatrix}$$

for any value of x_3 . ■

A square matrix A is **nonsingular** or **invertible** if its inverse A^{-1} such that $AA^{-1} = A^{-1}A = I$ exists. We can also define nonsingularity in terms of a matrix's **null space**, denoted $\text{null}(A)$, which is the set of points $\bar{v}$ such that $A\bar{v} = \bar{0}$. A matrix is nonsingular iff its null space has dimension 0.

For intuition, view square matrix A as a function that maps one point $\bar{u}$ to another $\bar{v} = A\bar{u}$. Suppose that A is not nonsingular so that $\text{null}(A)$ has dimension greater than 0: A sends more than one point to $\bar{0}$. In this case, one cannot construct an inverse function A^{-1} , as it would have to map $\bar{0}$ back to more than one point. It turns out that it is sufficient to consider only the points that A maps to $\bar{0}$: if A sends only $\bar{0}$ to $\bar{0}$, then the inverse A^{-1} exists. In other words, when A maps only $\bar{0}$ to $\bar{0}$, then it is a 1-to-1 map so that its inverse exists.

A 's inverse can be computed (on paper) with Gaussian elimination if it exists. Construct the augmented $n \times 2n$ -matrix $[A \mid I]$, and apply the elementary row operations to it until the left half is the identity matrix. The right half is then the inverse. To find just the k th column of A^{-1} , solve $A\bar{y} = \mathbf{e}_k$ by

Gaussian elimination for $\overline{y}$, rather than computing all of A^{-1} and extracting the k th column.

Example 8.4. To find the second column of the inverse of

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix},$$

solve

$$\underbrace{\begin{bmatrix} 3 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{e_2}.$$

Construct the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 0 \\ 1 & 0 & 1 & 1 \\ 2 & 2 & 1 & 0 \end{array} \right]$$

Apply the same row operations as in Example 8.2.

1. Add -2 times the first row and 4 times the second row to the third row:

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

2. Add -1 times the first row and 2 times the second row to the second row:

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 0 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

This augmented matrix is in triangular form.

Solving the resulting equations in reverse yields $x_1 = -3$, $x_2 = 1$, and $x_3 = 4$. Verify that $[-3 \ 1 \ 4]^T$ is indeed the second column of A^{-1} . ■

8.3 Linear Programs

Linear Inequalities

The **linear inequality**

$$G: A\overline{x} \leq \overline{b},$$

for $m \times n$ -matrix A , variable n -vector $\bar{x}$, and m -vector $\bar{b}$, compactly represents the $\Sigma_{\mathbb{Q}}$ -formula

$$G : \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i .$$

The subset of $\mathbb{Q}^n$ (or $\mathbb{R}^n$) that this inequality describes is called a **polyhedron**. Each member of this subset corresponds to one satisfying $T_{\mathbb{Q}}$ -interpretation of G . Exercises 8.3, 8.4, and 8.5 explore polyhedra in depth.

One important characteristic of the space defined by $A\bar{x} \leq \bar{b}$ is that it is **convex**. An n -dimensional space $S \subseteq \mathbb{R}^n$ is **convex** if for all pairs of points $\bar{v}_1, \bar{v}_2 \in S$,

$$\lambda \bar{v}_1 + (1 - \lambda) \bar{v}_2 \in S \quad \text{for } \lambda \in [0, 1] .$$

$A\bar{x} \leq \bar{b}$ defines a convex space. For suppose $A\bar{v}_1 \leq \bar{b}$ and $A\bar{v}_2 \leq \bar{b}$; then also

$$\lambda A\bar{v}_1 \leq \lambda \bar{b} \quad \text{and} \quad (1 - \lambda) A\bar{v}_2 \leq (1 - \lambda) \bar{b}$$

for $\lambda \in [0, 1]$. Summing each side of the inequalities yields

$$\begin{aligned} \lambda A\bar{v}_1 + (1 - \lambda) A\bar{v}_2 &\leq \lambda \bar{b} + (1 - \lambda) \bar{b} \\ A(\lambda \bar{v}_1 + (1 - \lambda) \bar{v}_2) &\leq \bar{b} \end{aligned}$$

as desired.

Consider when the $m \times n$ -matrix A is such that $m \geq n$. An n -vector $\bar{v}$ is a **vertex** of $A\bar{x} \leq \bar{b}$ if there is a nonsingular $n \times n$ -submatrix A_0 of A and corresponding n -subvector $\bar{b}_0$ of $\bar{b}$ such that $A_0\bar{v} = \bar{b}_0$. The rows in A_0 and $\bar{b}_0$ are the set of **defining constraints** of the vertex $\bar{v}$. In general, $\bar{v}$ may have multiple sets of defining constraints. Two vertices are **adjacent** if they have defining constraints that differ in only one constraint.

Intuitively, a vertex is an extremal point, such as the three vertices of a triangle. Adjacent vertices are connected by an edge of the space defined by $A\bar{x} \leq \bar{b}$; for example, each pair of vertices is adjacent in a triangle and connected by their common edge.

Linear Programs

The **linear optimization problem**, or **linear program**,

$$\begin{aligned} &\max \quad \bar{c}^T \bar{x} \\ &\text{subject to} \\ &\quad A\bar{x} \leq \bar{b} \end{aligned}$$

is solved by a point $\bar{v}^*$ that satisfies the **constraints** $A\bar{x} \leq \bar{b}$ and that maximizes the **objective function** $\bar{c}^T \bar{x}$. That is, $A\bar{v}^* \leq \bar{b}$, and $\bar{c}^T \bar{v}^*$ is maximal: $\bar{c}^T \bar{v}^* \geq \bar{c}^T \bar{u}$ for all $\bar{u}$ satisfying $A\bar{u} \leq \bar{b}$. If $A\bar{x} \leq \bar{b}$ is unsatisfiable, the maximum is $-\infty$ by convention. It is also possible that the maximum is unbounded, in which case the maximum is ∞ by convention.

Example 8.5. Consider the following linear program:

$$\begin{array}{ll}
 \max & \underbrace{\begin{bmatrix} 1 & 1 & -1 & -1 \end{bmatrix}}_{\bar{c}^T} \underbrace{\begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix}}_{\bar{x}} \\
 \text{subject to} & \underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix}}_{\bar{x}} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 3 \\ 2 \\ 2 \end{bmatrix}}_{\bar{b}}
 \end{array}$$

A is a 7×4 -matrix, $\bar{b}$ is a 7-vector, and $\bar{x}$ is a variable 4-vector representing the variables x, y, z_1, z_2 . The objective function is

$$(x - z_1) + (y - z_2) .$$

The constraints are equivalent to the $\Sigma_{\mathbb{Q}}$ -formula

$$\begin{array}{l}
 x \geq 0 \wedge y \geq 0 \wedge z_1 \geq 0 \wedge z_2 \geq 0 \\
 \wedge x + y \leq 3 \wedge x - z_1 \leq 2 \wedge y - z_2 \leq 2 .
 \end{array}$$

One vertex of the constraints is $\bar{v} = [2 \ 1 \ 0 \ 0]^T$. Why is it a vertex? Consider the submatrix A_0 of A consisting of rows 3, 4, 5, and 6; and the subvector $\bar{b}_0$ of $\bar{b}$ consisting of the same rows. A_0 is invertible. Additionally, $A_0\bar{v} = \bar{b}_0$:

$$\underbrace{\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}}_{A_0} \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\bar{v}} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 3 \\ 2 \end{bmatrix}}_{\bar{b}_0} .$$

Rows 3-6 are **equationally satisfied** in this case. Constraints 3, 4, 5, and 6 comprise the defining constraints of $\bar{v}$.

Another vertex is simply $[0 \ 0 \ 0 \ 0]^T$, for which the first four constraints are the defining constraints. ■

The following theorem is fundamental for solving linear programs. It asserts that the maximum value achieved by the objective function $\bar{c}^T\bar{x}$ over $\bar{x}$ satisfying $A\bar{x} \leq \bar{b}$ is equal to the minimum value achieved over the **dual optimization problem**.

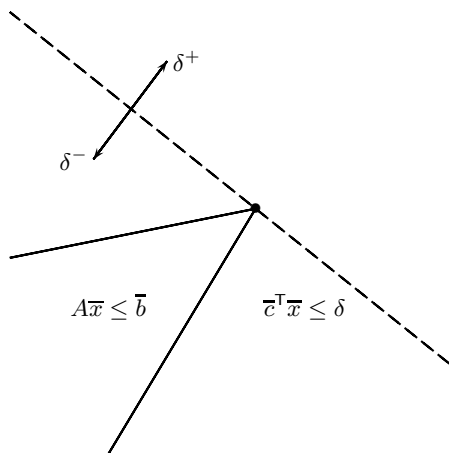


Fig. 8.1. Choosing δ

Theorem 8.6 (Duality Theorem of Linear Programming). *Consider $A \in \mathbb{Z}^{m \times n}$, $\bar{b} \in \mathbb{Z}^m$, and $\bar{c} \in \mathbb{Z}^n$. Then*

$$\max\{\bar{c}^T \bar{x} : A\bar{x} \leq \bar{b}\} = \min\{\bar{y}^T \bar{b} : \bar{y} \geq \bar{0} \wedge \bar{y}^T A = \bar{c}^T\}$$

if $A\bar{x} \leq \bar{b}$ is satisfiable.

By convention, when $A\bar{x} \leq \bar{b}$ is unsatisfiable, the maximum of the primal problem is $-\infty$, and the minimum of the dual form is ∞ .

The left and right optimization problems of the theorem are the **primal** and **dual** forms of the same problem. The theorem states that maximizing the function $\bar{c}^T \bar{x}$ over $A\bar{x} \leq \bar{b}$ is equivalent to minimizing the function $\bar{y}^T \bar{b}$ over all nonnegative $\bar{y}$ such that $\bar{y}^T A = \bar{c}^T$.

The theorem is actually surprisingly easy to visualize. In Figure 8.1, the region labeled $A\bar{x} \leq \bar{b}$ satisfies the inequality. The objective function $\bar{c}^T \bar{x}$ is represented by the dashed line. Its value increases in the direction of the arrow labeled δ^+ and decreases in the direction of the arrow labeled δ^- . Now, rather than asking for the maximum value of $\bar{c}^T \bar{x}$ over $\bar{x}$ satisfying $A\bar{x} \leq \bar{b}$, let us instead seek the minimal value δ such that

$$A\bar{x} \leq \bar{b} \Rightarrow \bar{c}^T \bar{x} \leq \delta.$$

In words, we seek the minimal δ such that $A\bar{x} \leq \bar{b}$ implies $\bar{c}^T \bar{x} \leq \delta$ or — visually — the region defined by $\bar{c}^T \bar{x} \leq \delta$ just covers the region defined by $A\bar{x} \leq \bar{b}$. Moving the dashed line of Figure 8.1 in the directions given by the arrows, we see that the best δ makes the dashed line just touch the $A\bar{x} \leq \bar{b}$ region (and at that “touched” point, $\bar{c}^T \bar{x}$ achieves its maximum, which is δ). Decreasing δ (δ^- direction) causes the implication to fail; increasing δ (δ^+ direction) causes the region defined by $\bar{c}^T \bar{x} \leq \delta$ to be unnecessarily large.

Therefore, we seek the right δ . Now consider multiplying the rows of $A\bar{x} \leq \bar{b}$ by nonnegative rationals and then summing the multiplied rows together. The resulting inequality is satisfied by any $\bar{x}$ satisfying $A\bar{x} \leq \bar{b}$; in other words, it is implied by $A\bar{x} \leq \bar{b}$. Mathematically, consider any nonnegative vector $\bar{y} \geq \bar{0}$; then

$$A\bar{x} \leq \bar{b} \Rightarrow \bar{y}^T A\bar{x} \leq \bar{y}^T \bar{b}.$$

Hence, to prove

$$A\bar{x} \leq \bar{b} \Rightarrow \bar{c}^T \bar{x} \leq \delta$$

for a fixed δ , find $\bar{y} \geq \bar{0}$ such that

$$\bar{y}^T A = \bar{c}^T \quad \text{and} \quad \bar{y}^T \bar{b} = \delta.$$

That is,

$$A\bar{x} \leq \bar{b} \Rightarrow \underbrace{\bar{c}^T}_{\bar{y}^T A} \bar{x} \leq \underbrace{\delta}_{\bar{y}^T \bar{b}}.$$

But we want to find a minimal δ such that the implication holds, not just prove it for a fixed δ . Thus, choose $\bar{y} \geq \bar{0}$ such that $\bar{y}^T A = \bar{c}^T$ and that minimizes $\bar{y}^T \bar{b}$. This equivalence between maximizing $\bar{c}^T \bar{x}$ and minimizing $\bar{y}^T \bar{b}$ is the one claimed by Theorem 8.6.

We refer the reader in **Bibliographic Remarks** to texts that contain the proof of this theorem.

$T_{\mathbb{Q}}$ -Satisfiability

Consider a generic $\Sigma_{\mathbb{Q}}$ -formula

$$F : \quad \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i \\ \wedge \quad \bigwedge_{i=1}^{\ell} \alpha_{i1}x_1 + \cdots + \alpha_{in}x_n < \beta_i$$

with both weak and strict inequalities. Equalities can be written as two inequalities. F is $T_{\mathbb{Q}}$ -equivalent to the $\Sigma_{\mathbb{Q}}$ -formula

$$F' : \quad \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i \\ \wedge \quad \bigwedge_{i=1}^{\ell} \alpha_{i1}x_1 + \cdots + \alpha_{in}x_n + x_{n+1} \leq \beta_i \\ \wedge \quad x_{n+1} > 0$$

with only weak inequalities except for $x_{n+1} > 0$. To decide the $T_{\mathbb{Q}}$ -satisfiability of F' , and thus of F , pose and solve the following linear program:

$$\begin{array}{ll} \max & x_{n+1} \\ \text{subject to} & \\ & \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i \\ & \bigwedge_{i=1}^{\ell} \alpha_{i1}x_1 + \cdots + \alpha_{in}x_n + x_{n+1} \leq \beta_i \end{array}$$

F' is $T_{\mathbb{Q}}$ -satisfiable iff the optimum is positive.

When F does not contain any strict inequality literals, the optimization problem is just a satisfiability problem because x_{n+1} is not introduced. This situation corresponds to a linear program with a constant objective value:

$$\begin{array}{ll} \max & 1 \\ \text{subject to} & \\ & \bigwedge_{i=1}^m a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i \end{array}$$

According to convention, the optimum is $-\infty$ iff the constraints are $T_{\mathbb{Q}}$ -unsatisfiable and 1 otherwise. This form of linear program is sometimes called a **linear feasibility problem**.

8.4 The Simplex Method

Consider the generic linear program

$$\begin{array}{ll} M : \max & \bar{c}^T \bar{x} \\ \text{subject to} & \\ & G : A\bar{x} \leq \bar{b} \end{array}$$

The **simplex method** solves the linear program in two main steps. In the first step, it obtains an initial vertex $\bar{v}_1$ of $A\bar{x} \leq \bar{b}$. In the second step, it iteratively traverses the vertices of $A\bar{x} \leq \bar{b}$, beginning at $\bar{v}_1$, in search of the vertex that maximizes the objective function. On each iteration of the second step, it determines if the current vertex $\bar{v}_i$ has a greater objective value than the vertices adjacent to $\bar{v}_i$. If not, it moves to one of the adjacent vertices with a greater objective value. If so, it halts and reports $\bar{v}_i$ as the optimum point with value $\bar{c}^T \bar{v}_i$.

$\bar{v}_i$ is a **local optimum** since its adjacent vertices have lesser objective values. But because the space defined by $A\bar{x} \leq \bar{b}$ is convex, $\bar{v}_i$ is also the **global optimum**: it is the highest value attained by any point that satisfies the constraints.

How does the simplex method find the initial vertex $\bar{v}_1$? In the first step, it constructs a new linear program

$$\begin{aligned} M_0 : \quad & \max \quad \bar{c}^T \bar{x} \\ & \text{subject to} \\ & G_0 : A' \bar{x} \leq \bar{b}' \end{aligned}$$

based only on $G : A\bar{x} \leq \bar{b}$ and for which, by construction, it has an initial vertex $\bar{v}'_1$. Moreover, by construction, M_0 achieves a certain optimal value v_G iff G is satisfiable; and if this optimal value is achieved, the point that achieves it is a vertex $\bar{v}_1$ of G . M_0 is solved by running the second step of the simplex method on M_0 itself, initialized with the known vertex $\bar{v}'_1$.

If the objective function of M is constant, then solving M_0 is the main step of the algorithm. G is satisfiable iff the optimum of M_0 is v_G . The second step is not applied to M in this case.

We now discuss the details of the simplex method.

8.4.1 From M to M_0

To find the initial vertex of M , the simplex method constructs and solves a new linear program M_0 . To that end, reformulate the constraints $G : A\bar{x} \leq \bar{b}$ of M so that they have the form

$$\bar{x} \geq 0, \quad A\bar{x} \leq \bar{b}$$

for new matrix A and vectors $\bar{x}$ and $\bar{b}$. A general technique is to introduce two nonnegative variables x_1, x_2 for each variable x and then to replace each instance of x with $x_1 - x_2$. Then we obtain a constraint system of the desired form that is $T_{\mathbb{Q}}$ -equisatisfiable to G .

Next, separate the new constraints $A\bar{x} \leq \bar{b}$ into two sets of inequalities

$$D_1 \bar{x} \leq \bar{g}_1 \quad \text{and} \quad D_2 \bar{x} \geq \bar{g}_2 \quad \text{for } \bar{g}_1 \geq \bar{0}, \bar{g}_2 > \bar{0}$$

according to the signs of the b_i . That is, if b_i is nonnegative, make row i of $A\bar{x} \leq \bar{b}$ a part of $D_1 \bar{x} \leq \bar{g}_1$; otherwise, multiply row i of $A\bar{x} \leq \bar{b}$ by -1 and make the result a part of $D_2 \bar{x} \geq \bar{g}_2$.

Pose the following optimization problem:

$$\begin{aligned} M_0 : \quad & \max \quad \bar{1}^T (D_2 \bar{x} - \bar{z}) \\ & \text{subject to} \\ & \bar{x}, \bar{z} \geq \bar{0} & (1) \\ & D_1 \bar{x} \leq \bar{g}_1 & (2) \\ & D_2 \bar{x} - \bar{z} \leq \bar{g}_2 & (3) \end{aligned} \tag{8.1}$$

The variable vector $\bar{z}$ has as many rows as D_2 . The objective function is the sum of the components of the vector $D_2 \bar{x} - \bar{z}$.

M_0 has several interesting characteristics:

1. The point $\bar{x} = \bar{0}$, $\bar{z} = \bar{0}$ satisfies constraints (1)-(3). It is a vertex.

- It satisfies constraints (2) and (3), for

$$D_1\bar{0} = \bar{0} \leq \bar{g}_1 \quad \text{and} \quad D_2\bar{0} - \bar{0} = \bar{0} \leq \bar{g}_2 .$$

The inequalities hold since $\bar{g}_1 \geq \bar{0}$ and $\bar{g}_2 > \bar{0}$ according to our construction of $\bar{g}_1$ and $\bar{g}_2$.

- It satisfies constraints (1). Indeed, constraints (1) are its defining constraints.
2. The optimum equals $v_G = \bar{1}^T \bar{g}_2$ iff G is $T_{\mathbb{Q}}$ -satisfiable.

- If the optimal value is $\bar{1}^T \bar{g}_2$ at optimal point $\bar{x}^*$, $\bar{z}^*$, then $\bar{x}^*$ satisfies G . For

$$D_1\bar{x}^* \leq \bar{g}_1$$

by (2); and

$$D_2\bar{x}^* - \bar{z}^* = \bar{g}_2$$

by constraint (3) of M_0 and that the optimal value is $\bar{1}^T \bar{g}_2$, so

$$D_2\bar{x}^* = \bar{g}_2 + \bar{z}^* \geq \bar{g}_2$$

by (1). By construction of D_1 , D_2 , $\bar{g}_1$, and $\bar{g}_2$, we thus know that $A\bar{x}^* \leq \bar{b}$, so G is satisfiable.

- If $\bar{x}^*$ satisfies G , then the optimal value is at least $\bar{1}^T \bar{g}_2$. We have

$$D_1\bar{x}^* \leq \bar{g}_1 , \quad D_2\bar{x}^* \geq \bar{g}_2 , \quad \text{and} \quad \bar{x}^* \geq \bar{0} .$$

Choose $\bar{z}^* = D_2\bar{x}^* - \bar{g}_2$. Then $\bar{z}^* \geq \bar{0}$ and $D_2\bar{x}^* - \bar{z}^* = \bar{g}_2$, so that all constraints of M_0 are satisfied. Additionally, $\bar{1}^T (D_2\bar{x}^* - \bar{z}^*) = \bar{1}^T \bar{g}_2$, so that the maximum of M_0 is at least $\bar{1}^T \bar{g}_2$. Constraint (3) limits the maximum to $\bar{1}^T \bar{g}_2$.

Because the optimum equals $\bar{1}^T \bar{g}_2$ iff G is $T_{\mathbb{Q}}$ -satisfiable, we need only solve the optimization problem M_0 to determine the satisfiability of G and to find a vertex of G if it is satisfiable. We already have a satisfying vertex $\bar{x} = \bar{0}$, $\bar{z} = \bar{0}$ of M_0 by characteristic 1.

Example 8.7. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$F : x + y \geq 1 \wedge x - y \geq -1 .$$

Because F has only weak inequality literals, the corresponding linear program has a constant objective function:

$$\begin{array}{ll} M : & \max \quad 1 \\ & \text{subject to} \\ & \left. \begin{array}{l} x + y \geq 1 \\ x - y \geq -1 \end{array} \right\} G \end{array}$$

Solving the corresponding linear program M_0 is sufficient to determine the $T_{\mathbb{Q}}$ -satisfiability of F .

To convert F to the form $\bar{x} \geq \bar{0} \wedge A\bar{x} \leq \bar{b}$, introduce nonnegative x_1, x_2 for x and y_1, y_2 for y :

$$F' : (x_1 - x_2) + (y_1 - y_2) \geq 1 \wedge (x_1 - x_2) - (y_1 - y_2) \geq -1 \\ \wedge x_1, x_2, y_1, y_2 \geq 0$$

F is $T_{\mathbb{Q}}$ -equisatisfiable to F' . In matrix form, the first two literals of F' are

$$\underbrace{\begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} \leq \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{\bar{b}}$$

Since $b_1 < 0$ and $b_2 > 0$, separating constraints yields

$$\underbrace{\begin{bmatrix} -1 & 1 & 1 & -1 \end{bmatrix}}_{D_1} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} \leq \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{\bar{g}_1} \quad \text{and} \quad \underbrace{\begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}}_{D_2} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} \geq \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{\bar{g}_2}.$$

D_2 has only one row, so $\bar{z} = [z]$. According to (8.1), pose the following optimization problem:

$$\begin{aligned} & \mathbf{max} \quad \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} - [z] \\ & \mathbf{subject\ to} \\ & \quad x_1, x_2, y_1, y_2, z \geq 0 \\ & \quad \begin{bmatrix} -1 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} \leq [1] \\ & \quad \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} - [z] \leq [1] \end{aligned}$$

F is $T_{\mathbb{Q}}$ -satisfiable iff the optimum is $v_G = \bar{1}^T \bar{g}_2 = 1$.

We know that the point $[x_1 \ x_2 \ y_1 \ y_2 \ z]^T = [0 \ 0 \ 0 \ 0 \ 0]^T$ is a vertex. It satisfies all constraints and has defining constraints $x_1, x_2, y_1, y_2, z \geq 0$. ■

The simplex method requires M_0 to be in standard form

$$\begin{array}{ll} \max & \bar{c}^\top \bar{y} \\ \text{subject to} & A\bar{y} \leq \bar{b} \end{array}$$

Let

$$\bar{y} = \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}.$$

Since the objective function of M_0 is

$$\bar{1}^\top (D_2 \bar{x} - \bar{z}) = \bar{1}^\top \underbrace{\begin{bmatrix} D_2 & -I \end{bmatrix} \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}}_{\bar{y}},$$

construct vector $\bar{c}$ as follows:

$$\bar{c}^\top = \bar{1}^\top \begin{bmatrix} D_2 & -I \end{bmatrix}.$$

The constraints have the form

$$\begin{bmatrix} -I & -I \\ D_1 & -I \\ D_2 & -I \end{bmatrix} \underbrace{\begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}}_{\bar{y}} \leq \begin{bmatrix} \bar{0} \\ \bar{0} \\ \bar{g}_1 \\ \bar{g}_2 \end{bmatrix},$$

where blank regions of the matrix are filled with 0s. Hence, M_0 of (8.1) in standard form is written

$$\begin{array}{ll} M_0 : \max & \underbrace{\bar{1}^\top \begin{bmatrix} D_2 & -I \end{bmatrix}}_{\bar{c}^\top} \underbrace{\begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}}_{\bar{y}} \\ & \text{subject to} \\ & \underbrace{\begin{bmatrix} -I & -I \\ D_1 & -I \\ D_2 & -I \end{bmatrix}}_A \underbrace{\begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}}_{\bar{y}} \leq \underbrace{\begin{bmatrix} \bar{0} \\ \bar{0} \\ \bar{g}_1 \\ \bar{g}_2 \end{bmatrix}}_{\bar{b}} \end{array} \quad (8.2)$$

Example 8.8. According to (8.2), rewrite the optimization problem of Example 8.7 as follows:

$$\begin{array}{ll}
\max & \underbrace{\begin{bmatrix} 1 & -1 & 1 & -1 & -1 \end{bmatrix}}_{\bar{c}^\top} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ z \end{bmatrix} \\
\text{subject to} & \underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ -1 & 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & -1 & -1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ z \end{bmatrix} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}}_{\bar{b}}
\end{array}$$

■

8.4.2 Vertex Traversal

Assume that we have a generic optimization problem

$$\begin{array}{ll}
\max & \bar{c}^\top \bar{x} \\
\text{subject to} & A\bar{x} \leq \bar{b}
\end{array} \tag{8.3}$$

for which we have a satisfying vertex $\bar{v}_i$. The simplex method traverses vertices of $A\bar{x} \leq \bar{b}$ to find the vertex $\bar{v}^*$ that maximizes $\bar{c}^\top \bar{x}$. In particular, one iteration of this step seeks a vertex $\bar{v}_{i+1}$ adjacent to $\bar{v}_i$ such that $\bar{c}^\top \bar{v}_{i+1} > \bar{c}^\top \bar{v}_i$.

For the optimization problem M_0 of (8.2), the point $\bar{x} = \bar{0}$, $\bar{z} = \bar{0}$, which satisfies the constraints of M_0 , is the initial vertex $\bar{v}_1$.

Example 8.9. The point $\bar{v}_1 = [x_1 \ x_2 \ y_1 \ y_2 \ z]^\top = [0 \ 0 \ 0 \ 0 \ 0]^\top$ is a vertex of the constraints of the optimization problem of Example 8.8. ■

Construction of $\bar{u}$

To begin the i th iteration, we use the vertex $\bar{v}_i$ to construct a vector $\bar{u}$ such that $\bar{u}^\top A = \bar{c}^\top$. If $\bar{u} \geq \bar{0}$ then the Duality Theorem (Theorem 8.6) implies that $\bar{v}_i$ is optimal, as we discuss below, and the process terminates. However, in all but the final iteration, $\bar{u} \not\geq \bar{0}$: at least one row of $\bar{u}$ is negative.

To construct $\bar{u}$, choose one of its sets of defining constraints: choose a $n \times n$ nonsingular submatrix A_i of A with corresponding rows $\bar{b}_i$ such that

$$A_i \bar{v}_i = \bar{b}_i. \tag{8.4}$$

Let R be the indices of the rows of A in A_i . Such a subset of constraints exists because $\bar{v}_i$ is a vertex.

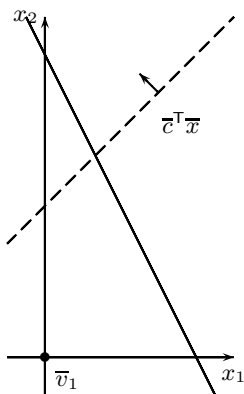


Fig. 8.2. Illustration of Example 8.10

Second, solve $\bar{u}^T A = \bar{c}^T$ for $\bar{u}$: solve

$$A_i^T \bar{u}_i = \bar{c} \quad (8.5)$$

for $\bar{u}_i$ (which has a solution because A_i is nonsingular), and let $\bar{u}$ be $\bar{u}_i$ for indices in R with 0s added for rows with indices not in R . Thus,

$$\bar{u}^T A = \bar{c}^T. \quad (8.6)$$

Example 8.10. Consider the optimization problem of the form (8.3)

$$\begin{array}{ll} \max & \underbrace{[-1 \ 1]}_{\bar{c}^T} \bar{x} \\ \text{subject to} & \underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 2 & 1 \end{bmatrix}}_A \bar{x} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}}_{\bar{b}} \end{array}$$

for which we know that $\bar{v}_1 = [0 \ 0]^T$ is a vertex.

The problem and initial vertex is visualized in Figure 8.2. The solid lines represent the constraints of the problem, and the set of satisfying points corresponds to the interior of the triangle. The dashed line indicates $\bar{c}^T \bar{x}$; the arrow points in the direction of increasing value.

Given vertex $\bar{v}_1 = [0 \ 0]^T$, the first two constraints are the defining constraints of $\bar{v}_1$, so choose $R = [1; 2]$:

$$A_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad \bar{b}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

From (8.5), solving

$$\underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}}_{A_1^\top} \bar{u}_1 = \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{\bar{c}}$$

for $\bar{u}_1$ (via, for example, Gaussian elimination; or by observing that $A_1^\top = -I$ and that $-I\bar{u}_1 = \bar{c}$ implies that $\bar{u}_1 = -\bar{c}$) yields $\bar{u}_1 = [1 \ -1]^\top$. Adding 0s for rows not in R produces

$$\bar{u} = [1 \ -1 \ 0]^\top,$$

where the first two elements are from $\bar{u}_1$. Check that this $\bar{u}$ satisfies $\bar{u}^\top A = \bar{c}^\top$ of (8.6) as desired. ■

Example 8.11. Continuing from Examples 8.8 and 8.9, choose the first five rows of A and $\bar{b}$ ($R = [1; 2; 3; 4; 5]$) since

$$\underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}}_{A_1} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{v}_1} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{b}_1}$$

as desired from (8.4). From (8.5), solving

$$\underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}}_{A_1^\top} \bar{u}_1 = \underbrace{\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ -1 \end{bmatrix}}_{\bar{c}}$$

yields

$$\bar{u}_1^\top = [-1 \ 1 \ -1 \ 1 \ 1].$$

Then

$$\bar{u} = [-1 \ 1 \ -1 \ 1 \ 1 \ 0 \ 0]^\top,$$

where the first five elements are from $\bar{u}_1$. Check that this $\bar{u}$ satisfies $\bar{u}^\top A = \bar{c}^\top$ of (8.6) as desired. ■

There are two cases to consider: either $\bar{u} \geq \bar{0}$ or $\bar{u} \not\geq \bar{0}$.

Case 1: $\bar{u} \geq \bar{0}$

We prove that in this case, $\bar{v}_i$ is actually the optimal point with optimal value $\bar{c}^\top \bar{v}_i$. The crux of the argument is the Duality Theorem (Theorem 8.6).

From equation (8.6), we have

$$\bar{c}^\top \bar{v}_i = \bar{u}^\top A \bar{v}_i .$$

We claim next that

$$\bar{u}^\top A \bar{v}_i = \bar{u}^\top \bar{b} . \quad (8.7)$$

First, from (8.4)

$$A_i \bar{v}_i = \bar{b}_i \quad \text{implies} \quad \bar{u}_i^\top A_i \bar{v}_i = \bar{u}_i^\top \bar{b}_i$$

so that equation (8.7) holds at rows R . For rows $j \notin R$, we know that $u_j = 0$ by construction, so that both

$$(\bar{u}^\top A \bar{v}_i)_j = 0 \quad \text{and} \quad (\bar{u}^\top \bar{b})_j = 0 ,$$

proving equation (8.7). Reasoning further,

$$\bar{u}^\top \bar{b} \geq \min\{\bar{y}^\top \bar{b} : \bar{y} \geq \bar{0} \wedge \bar{y}^\top A = \bar{c}^\top\}$$

since $\bar{u}$ is a member of the set by (8.6) and the case $\bar{u} \geq \bar{0}$. By duality (Theorem 8.6),

$$\min\{\bar{y}^\top \bar{b} : \bar{y} \geq \bar{0} \wedge \bar{y}^\top A = \bar{c}^\top\} = \max\{\bar{c}^\top \bar{x} : A \bar{x} \leq \bar{b}\} .$$

In summary, we have by (8.6), (8.7), and Theorem 8.6,

$$\begin{aligned} \bar{c}^\top \bar{v}_i &= \bar{u}^\top A \bar{v}_i = \bar{u}^\top \bar{b} \geq \\ &\min\{\bar{y}^\top \bar{b} : \bar{y} \geq \bar{0} \wedge \bar{y}^\top A = \bar{c}^\top\} = \max\{\bar{c}^\top \bar{x} : A \bar{x} \leq \bar{b}\} , \end{aligned}$$

which proves that $\bar{v}_i$ is actually the optimal point with optimal value $\bar{c}^\top \bar{v}_i$.

Figure 8.3 illustrates this case: the vertex $\bar{v}_i$ maximizes $\bar{c}^\top \bar{x}$. The dashed line illustrates the objective function $\bar{c}^\top \bar{x}$. Moving upward relative to it increases its value. In this illustration, $\bar{c}^\top \bar{x}$ cannot be increased without leaving the region defined by the constraints.

Case 2: $\bar{u} \not\geq \bar{0}$

In this case, $\bar{v}_i$ is not the optimal point. Thus, we need to move along an edge $\bar{y}$ to an adjacent vertex to increase the value of the objective function. In moving to an adjacent vertex, we swap one of the defining constraints of $\bar{v}_i$ for another constraint to form the defining constraints of $\bar{v}_{i+1}$.

In this second case, there exists some $u_k < 0$. Let k be the lowest index of $\bar{u}$ such that $u_k < 0$ (it must be one of the indices of R since for all other indices ℓ , $u_\ell = 0$). Let k' be the index of the row of $\bar{u}_i$ and A_i corresponding to row k of $\bar{u}$ and A_i .

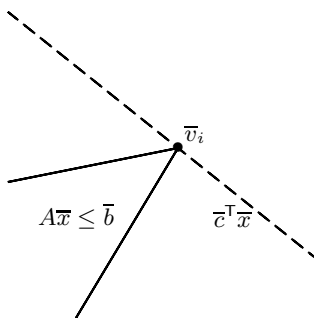


Fig. 8.3. Case 1

Construction of $\bar{y}$

Having fixed the indices k (row k of A and $\bar{b}$) and k' (the corresponding row k' of A_i and $\bar{b}_i$) of the offending constraint, we seek a direction along which to travel away from vertex $\bar{v}_i$ and, in particular, away from the plane that defines the k th constraint.

Define $\bar{y}$ to be the k' th column of $-A_i^{-1}$. To find $\bar{y}$, solve

$$A_i \bar{y} = -\mathbf{e}_{k'} \quad (8.8)$$

(where, recall, $\mathbf{e}_{k'}$ is the k' th unit vector, which consists of 0s except in position k') for $\bar{y}$. Thus,

$$\bar{a}_\ell \bar{y} = 0 \quad \text{for every row } \bar{a}_\ell \text{ of } A_i \text{ except the } k'\text{th row}$$

and

$$\bar{a}_{k'} \bar{y} = -1 \quad \text{for the } k'\text{th row } \bar{a}_{k'} \text{ of } A_i.$$

The vector $\bar{y}$ provides the direction along which to move to the next vertex. For all rows $\bar{a}_\ell$ but the k' th row of A_i , $\bar{a}_\ell \bar{y} = 0$ implies that moving in direction $\bar{y}$ stays on the boundary of the constraints (*i.e.*, both $\bar{a}_\ell \bar{v}_i = \bar{b}$ and, in the next step, $\bar{a}_\ell \bar{v}_{i+1} = \bar{b}$), so $\bar{a}_\ell$ will be a row in A_{i+1} . However, moving along $\bar{y}$ moves inward from the boundary of the k' th constraint because $\bar{a}_{k'} \bar{y} = -1$. This change is desirable, as this constraint is keeping $\bar{u}$ from being nonnegative.

Example 8.12. Let us examine

$$\bar{u}_1 = [1 \ -1]^\top \quad \text{and} \quad \bar{u} = [1 \ -1 \ 0]^\top$$

of Example 8.10. Since the second row of $\bar{u}$ is -1 , we are in Case 2 with $k = 2$, corresponding to row $k' = 2$ of $\bar{u}_1$. Let $\bar{y}$ be the second column of $-A_1^{-1}$: solve

$$\underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}}_{A_1} \bar{y} = \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_{-\mathbf{e}_2}$$

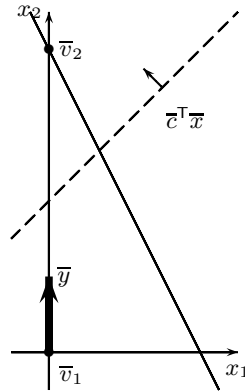


Fig. 8.4. Illustration of Example 8.12

for $\bar{y}$, yielding $\bar{y} = [0 \ 1]^\top$.

This $\bar{y}$ is visualized in Figure 8.4 by the dark solid arrow that points up from $\bar{v}_1$. The vertical and horizontal lines are the defining constraints of $\bar{v}_1$; in moving in the direction $\bar{y}$, we keep the vertical constraint for the next vertex $\bar{v}_2$ but drop the horizontal constraint. The diagonal constraint will become the second of $\bar{v}_2$'s defining constraints. ■

Example 8.13. Let us examine

$$\bar{u}_1 = [-1 \ 1 \ -1 \ 1 \ 1]^\top \quad \text{and} \quad \bar{u} = [-1 \ 1 \ -1 \ 1 \ 1 \ 0 \ 0]^\top$$

of Example 8.11. Since the first row of $\bar{u}$ is -1 , $k = 1$, corresponding to row $k' = 1$ of $\bar{u}_1$. Thus, solve

$$\underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}}_{A_1} \bar{y} = \underbrace{\begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{-\mathbf{e}_1}$$

for $\bar{y}$, yielding $\bar{y} = [1 \ 0 \ 0 \ 0 \ 0]^\top$. ■

Again we have two cases to consider: either the optimum is bounded (**Case 2(a)**) or it is unbounded (**Case 2(b)**).

Case 2(a): Optimum is Bounded

In this case, we move along the edge $\bar{y}$ to a better vertex $\bar{v}_{i+1}$, according to the objective function $\bar{c}^\top \bar{x}$. However, there is a set of rows of A with indices

S such that for $\ell \in S$, $\bar{a}_\ell \bar{y} > 0$. For these constraints, moving in the direction $\bar{y}$ actually moves toward leaving the satisfying region. These constraints limit how far in direction $\bar{y}$ we can move. For example, in Figure 8.4, the diagonal constraint limits how far we can move in direction $\bar{y}$.

Construction of λ_i , $\bar{v}_{i+1}$

We want to solve for the greatest $\lambda_i \geq 0$ such that

$$A(\bar{v}_i + \lambda_i \bar{y}) \leq \bar{b} . \quad (8.9)$$

Thus, choose $\lambda_i > 0$ such that for some row $\bar{a}_\ell$, $\ell \in S$,

$$\bar{a}_\ell(\bar{v}_i + \lambda_i \bar{y}) = b_\ell \quad (8.10)$$

and for all other rows $m \in S \setminus \{\ell\}$,

$$\bar{a}_m(\bar{v}_i + \lambda_i \bar{y}) \leq b_m . \quad (8.11)$$

Set

$$\bar{v}_{i+1} \stackrel{\text{def}}{=} \bar{v}_i + \lambda_i \bar{y} . \quad (8.12)$$

Finally, we construct the defining constraints of $\bar{v}_{i+1}$ for the next iteration. Construct submatrix A_{i+1} of A from A_i : replace row $\bar{a}_{k'}$ of A_i with row $\bar{a}_\ell$ of A . Choose the corresponding rows of $\bar{b}$ for $\bar{b}_{i+1}$. Row ℓ and the rows carried over from this iteration comprise the defining constraints of $\bar{v}_{i+1}$.

Example 8.14. Continuing from Example 8.12, choose λ_1 such that

$$A(\bar{v}_1 + \lambda_1 \bar{y}) \leq \bar{b} ,$$

specifically

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 2 & 1 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \leq \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} ,$$

and one constraint is equationally satisfied. Thus, choose $\lambda_1 = 2$.

Notice that the first and third constraints are equationally satisfied. The first row is already included in R ; hence, focus on the third row ($\ell = 3$). From (8.12), define v_{i+1} for the next iteration:

$$\bar{v}_2 = \bar{v}_1 + \lambda_1 \bar{y} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} .$$

This vertex is visualized in Figure 8.4. Choosing $R = [1; 3]$ and replacing the second row of A_1 and $\bar{b}_1$ with the third row of $A\bar{x} \leq \bar{b}$ yields

$$A_2 = \begin{bmatrix} -1 & 0 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad \bar{b}_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

In Figure 8.4, the rows of A_2 and $\bar{b}_2$ correspond to the vertical and diagonal constraints, respectively, which are the defining constraints of $\bar{v}_2$.

In the next iteration, solving $A_2^\top \bar{u}_2 = \bar{c}$ yields $\bar{u}_2 = [3 \ 1]^\top$. Adding 0s for rows not in R produces

$$\bar{u} = [3 \ 0 \ 1]^\top.$$

Since $\bar{u} \geq \bar{0}$, we are in Case 1. The maximum is

$$\bar{c}^\top \bar{v}_2 = [-1 \ 1] \begin{bmatrix} 0 \\ 2 \end{bmatrix} = 2$$

at vertex $\bar{v}_2$. ■

This move from vertex $\bar{v}_i$ to vertex $\bar{v}_{i+1}$ makes progress. For

$$\begin{aligned} \bar{c}^\top \bar{v}_{i+1} &= \bar{c}^\top (\bar{v}_i + \lambda_i \bar{y}) \\ &= \bar{c}^\top \bar{v}_i + \lambda_i \bar{c}^\top \bar{y} \\ &= \bar{c}^\top \bar{v}_i + \lambda_i (-u_k) \\ &> \bar{c}^\top \bar{v}_i. \end{aligned}$$

When considering line 3, recall from (8.5) of the construction of $\bar{u}$ that

$$\bar{u}_i^\top = \bar{c}^\top A_i^{-1}$$

and that the k 'th column of A_i^{-1} is $-\bar{y}$ by (8.8). Hence, $\bar{c}^\top \bar{y} = -u_k$.

Moreover, since there are only a finite number of vertices to examine, **Case 1** (or, in the general case, possibly **Case 2(b)**) eventually occurs.

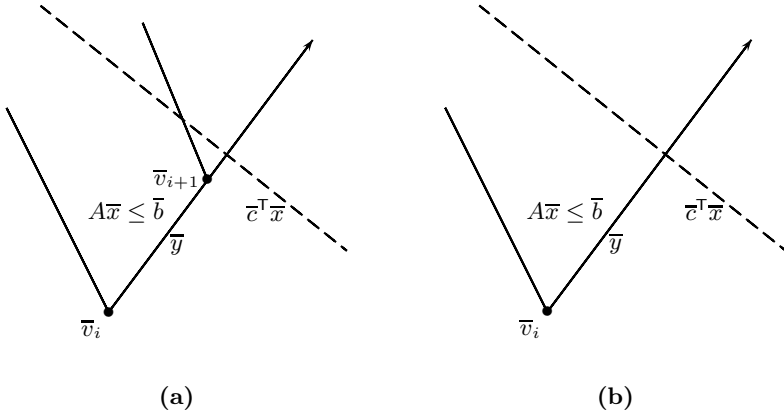
Figure 8.5(a) illustrates this case: vertex $\bar{v}_{i+1}$ is discovered by moving along ray $\bar{y}$ as far as possible without violating the constraints. Moreover $\bar{c}^\top \bar{v}_{i+1}$ is greater than $\bar{c}^\top \bar{v}_i$.

It is possible that $\lambda_i = 0$ when the vertex $\bar{v}_i$ has multiple sets of defining constraints. However, because there are only a finite number of constraints, the number of defining constraint sets is also finite. Preventing repetitions of the matrix A_i across iterations guarantees that the algorithm still halts in this case.

Case 2(b): Optimum is Unbounded

In this case, $A\bar{y} \leq \bar{0}$ and the optimum is unbounded. Since $A\bar{y} \leq \bar{0}$, for all $\lambda \geq 0$,

$$A(\bar{v}_i + \lambda \bar{y}) = \underbrace{A\bar{v}_i}_{\leq \bar{b}} + \lambda \underbrace{A\bar{y}}_{\leq \bar{0}} \leq \bar{b},$$

**Fig. 8.5.** Case 2

so all $\bar{v}_i + \lambda \bar{y}$ satisfy the constraints. In other words, moving in direction $\bar{y}$ either moves along the boundary of a constraint ($\bar{a}_i \bar{y} = 0$) or moves inward ($\bar{a}_i \bar{y} < 0$). Moreover,

$$\begin{aligned} \bar{c}^T (\bar{v}_i + \lambda \bar{y}) &= \bar{c}^T \bar{v}_i + \lambda \bar{c}^T \bar{y} \\ &= \bar{c}^T \bar{v}_i + \lambda \bar{u}^T A \bar{y} \\ &= \bar{c}^T \bar{v}_i + \underbrace{-\lambda}_{<0} u_k, \end{aligned}$$

which grows with λ (recall $\bar{c}^T = \bar{u}^T A$, and $A \bar{y} = [0 \cdots 0 -1 0 \cdots 0]^T$ by selection of $\bar{y}$). Thus, the maximum is unbounded. Figure 8.5(b) illustrates this case: all points along the ray labeled $\bar{y}$ satisfy the constraints, and moving along the ray increases $\bar{c}^T \bar{x}$ without bound.

Example 8.15. Continuing with $\bar{y} = [1 \ 0 \ 0 \ 0 \ 0]^T$ from Example 8.13, compute $A \bar{y}$

$$\underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ -1 & 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & -1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{y}} = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

to find that $S = [7]$ since $\bar{a}_7 \bar{y} = 1 > 0$. Thus, according to (8.10), examine the seventh row of the constraints and choose the greatest λ_1 such that

$$\underbrace{[1 \ -1 \ 1 \ -1 \ -1]}_{\bar{a}_7}(\bar{v}_1 + \lambda_1 \bar{y}) = [1 \ -1 \ 1 \ -1 \ -1] \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right) = 1 ;$$

that is, choose $\lambda_1 = 1$. By (8.12), define

$$\bar{v}_2 = \bar{v}_1 + \lambda_1 \bar{y} = [1 \ 0 \ 0 \ 0 \ 0]^\top .$$

Form A_2 by replacing the first row ($k' = 1$) of A_1 with the seventh row ($\ell = 7$) of A :

$$A_2 = \begin{bmatrix} 1 & -1 & 1 & -1 & -1 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix} .$$

This move to vertex $\bar{v}_2$ makes progress:

$$\bar{c}^\top \bar{v}_1 = 0 < \bar{c}^\top \bar{v}_2 = 1 .$$

Now $R = [7; 2; 3; 4; 5]$; that is, A_2 is the submatrix of A constructed from rows 7, 2, 3, 4, and 5. Solve $A_2^\top \bar{u}_2 = \bar{c}$ for $\bar{u}_2$, yielding $\bar{u}_2 = [1 \ 0 \ 0 \ 0 \ 0]^\top$. Since $\bar{u}_2 \geq 0$, we are in **Case 1**: we have found an optimum point $\bar{v}_2$ with optimal value 1.

Recall from Example 8.7 that $v_G = \bar{1}^\top \bar{g}_2 = 1$. The equality of the optimum and v_G implies that

$$F : x + y \geq 1 \ \wedge \ x - y \geq -1$$

is $T_{\mathbb{Q}}$ -satisfiable. In particular, extract from

$$\begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ z \end{bmatrix} = \bar{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

the assignment

$$x = x_1 - x_2 = 1 - 0 = 1 \quad \text{and} \quad y = y_1 - y_2 = 0 - 0 = 0 ,$$

which indeed satisfies F . ■

Example 8.16. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$F : x \geq 0 \ \wedge \ y \geq 0 \ \wedge \ x \geq 2 \ \wedge \ y \geq 2 \ \wedge \ x + y \leq 3 ,$$

or, in matrix form,

$$F : \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \\ -2 \\ -2 \\ 3 \end{bmatrix}.$$

Is F $T_{\mathbb{Q}}$ -satisfiable? Because F has only weak inequality literals, the corresponding linear program has a constant objective function:

$$\begin{aligned} M : \quad & \mathbf{max} \quad 1 \\ & \mathbf{subject \ to} \\ & G : \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \\ -2 \\ -2 \\ 3 \end{bmatrix} \end{aligned}$$

Solving the corresponding linear program M_0 is sufficient to determine the $T_{\mathbb{Q}}$ -satisfiability of F .

Construction of M_0

Because x and y are already constrained to be nonnegative, we do not need to introduce new x_1, x_2, y_1, y_2 . Rewrite the final three literals of F as two sets of constraints:

$$\underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{D_1} \begin{bmatrix} x \\ y \end{bmatrix} \leq \underbrace{[3]}_{\bar{g}_1} \quad \text{and} \quad \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{D_2} \begin{bmatrix} x \\ y \end{bmatrix} \geq \underbrace{\begin{bmatrix} 2 \\ 2 \end{bmatrix}}_{\bar{g}_2}$$

so that $\bar{g}_1 \geq 0$ and $\bar{g}_2 > 0$. Then pose the following optimization problem of form (8.2):

$$\begin{aligned} M_0 : \quad & \mathbf{max} \quad \underbrace{\begin{bmatrix} 1 & 1 & -1 & -1 \end{bmatrix}}_{\bar{c}^T} \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} \\ & \mathbf{subject \ to} \\ & \underbrace{\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}_A \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 3 \\ 2 \\ 2 \end{bmatrix}}_{\bar{b}} \end{aligned}$$

in which $\bar{c}$ is computed according to

$$\begin{aligned}\bar{c}^\top \bar{x} &= \bar{1}^\top [D_2 \ -I] \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} = [1 \ 1] \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} \\ &= \underbrace{[1 \ 1 \ -1 \ -1]}_{\bar{c}^\top} \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} .\end{aligned}$$

Use the initial vertex

$$\bar{v}_1 = \begin{bmatrix} x \\ y \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

to start the vertex traversal.

F is satisfiable iff the optimal value is equal to

$$v_G = \bar{1}^\top \bar{g}_2 = [1 \ 1] \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 4 .$$

We apply the simplex method to find the optimum.

Iteration 1

Choose rows $R = [1; 2; 3; 4]$ of A and $\bar{b}$ to form

$$A_1 = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \bar{b}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} .$$

This submatrix and subvector correspond to the defining constraints of vertex $\bar{v}_1$: $A_1 \bar{v}_1 = \bar{b}_1$.

Recall that by the Duality Theorem (Theorem 8.6), our goal is to find a vector $\bar{u}$ such that $\bar{u} \geq \bar{0}$ and $\bar{u}^\top A = \bar{c}^\top$. We use A_1 to solve for a $\bar{u}$ satisfying $\bar{u}^\top A = \bar{c}^\top$ and, through iteratively exploring a sequence of vertices, eventually find a $\bar{u}$ that also satisfies $\bar{u} \geq \bar{0}$.

Therefore, we solve for $\bar{u}$. By (8.5), solve $A_1^\top \bar{u}_1 = \bar{c}$ to yield $\bar{u}_1 = [-1 \ -1 \ 1 \ 1]^\top$. Adding 0s for the rows not in R produces $\bar{u}$:

$$\bar{u} = [-1 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0]^\top ,$$

which satisfies $\bar{u}^\top A = \bar{c}^\top$ (8.6) as desired.

Since $u_1, u_2 < 0$, we are in **Case 2** with $k = k' = 1$. We have not yet found a $\bar{u} \geq \bar{0}$, so we must continue searching. In particular, constraints 1 and 2 are causing a problem (as indicated by $u_1, u_2 < 0$), so we must find a direction $\bar{y}$ and distance λ_1 to travel to a better vertex whose defining constraints do not include constraint 1. The direction is given by the first column of $-A_1^{-1}$: by (8.8), solve $A_1 \bar{y} = -\mathbf{e}_1$ to yield $\bar{y} = [1 \ 0 \ 0 \ 0]^T$.

It remains to find the distance λ_1 to travel along $\bar{y}$. Computing $A\bar{y}$ shows that $S = [5; 6]$: the fifth and sixth rows $\bar{a}$ of A are such that $\bar{a}\bar{y} > 0$. This situation means that traveling along $\bar{y}$ too far will result in making one or both of constraints 5 and 6 unsatisfied. The other constraints will continue to be satisfied independent of λ_1 .

Therefore, by (8.9), choose the largest λ_1 such that

$$A(\bar{v}_1 + \lambda_1 \bar{y}) \leq \bar{b}.$$

Focusing on the fifth and sixth rows of A (since $S = [5; 6]$), choose the largest λ_1 such that

$$\underbrace{\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}}_{\text{rows 5,6 of } A} \left(\underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{v}_1} + \lambda_1 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{y}} \right) \leq \underbrace{\begin{bmatrix} 3 \\ 2 \end{bmatrix}}_{\text{rows 5,6 of } \bar{b}}$$

according to (8.9). Namely, choose $\lambda_1 = 2$ (and $\ell = 6$).

We now have a direction ($\bar{y}$) and a distance (λ_1) to travel away from $\bar{v}_1$. By (8.12), define

$$\bar{v}_2 = \bar{v}_1 + \lambda_1 \bar{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

For the next iteration, replace the first row of A_1 (since $k' = 1$) with the sixth row of A (since $\ell = 6$) to produce

$$A_2 = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \bar{b}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Have we made progress by moving to vertex $\bar{v}_2$? Yes, for

$$\bar{c}^T \bar{v}_1 = 0 < 2 = \bar{c}^T \bar{v}_2.$$

The objective function has increased from 0 to 2.

Iteration 2

Now $R = [6; 2; 3; 4]$. Solve $A_2^T \bar{u}_2 = \bar{c}$ to yield $\bar{u}_2 = [1 \ -1 \ 0 \ 1]^T$ for rows $[6; 2; 3; 4]$. Then filling in 0s for the other rows of A produces:

$$\bar{u} = [0 \ -1 \ 0 \ 1 \ 0 \ 1 \ 0]^T .$$

row: 2 3 4 6

Note that $\bar{u}$ maintains the order of rows, while $\bar{u}_2$ is ordered according to R . $u_2 < 0$, so $k = 2$, which corresponds to row $k' = 2$ of $\bar{u}_2$. According to **Case 2**, let $\bar{y}$ be the second column of $-A_2^{-1}$: solve $A_2 \bar{y} = -\mathbf{e}_2$ to yield $\bar{y} = [0 \ 1 \ 0 \ 0]^T$. Then the fifth and seventh rows $\bar{a}$ of A are such that $\bar{a}\bar{y} > 0$ so that $S = [5; 7]$. Focusing on the fifth and seventh rows of A , choose the largest λ_2 such that

$$\underbrace{\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}_{\text{rows 5,7 of } A} \left(\underbrace{\begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\bar{v}_2} + \lambda_2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\bar{b}} \right) \leq \underbrace{\begin{bmatrix} 3 \\ 2 \end{bmatrix}}_{\text{rows 5,7 of } \bar{b}} .$$

Choose $\lambda_2 = 1$ (and $\ell = 5$). Then

$$\bar{v}_3 = \bar{v}_2 + \lambda_2 \bar{y} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} .$$

Replace the second row of A_2 (since $k' = 2$) with the fifth row of A (since $\ell = 5$) to produce

$$A_3 = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \bar{b}_3 = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 0 \end{bmatrix} .$$

Have we made progress? Yes, for

$$\bar{c}^T \bar{v}_1 = 0 < \bar{c}^T \bar{v}_2 = 2 < \bar{c}^T \bar{v}_3 = 3 .$$

The objective function has increased from 2 to 3.

Iteration 3

Now $R = [6; 5; 3; 4]$. Solve $A_3^T \bar{u}_3 = \bar{c}$, yielding $\bar{u}_3 = [0 \ 1 \ 1 \ 1]^T$. Because $\bar{u}_3 \geq \bar{0}$, we are in **Case 1**: $\bar{v}_3$ is the optimum with objective value

$$\bar{c}^T \bar{v}_3 = [1 \ 1 \ -1 \ -1] \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = 3 .$$

The optimal value of the constructed optimization problem is 3, which is less than the required $v_G = 4$. The constraints G of the linear program M are unsatisfiable, and thus F itself is $T_{\mathbb{Q}}$ -unsatisfiable. ■

8.4.3 ★Complexity

Theorem 8.17. *$T_{\mathbb{Q}}$ -satisfiability of conjunctive quantifier-free $\Sigma_{\mathbb{Q}}$ -formulae is weakly polynomial-time decidable.*

An algorithm is weakly polynomial-time if it runs in time polynomial in the actual values of the input, not just the size of the input. **Bibliographic Remarks** discusses algorithms that achieve this performance.

While efficient in practice, the simplex method actually has poor worst-case behavior for known fixed pivot rules. A **pivot rule** determines which adjacent vertex to visit next. In our presentation, no rule is fixed: the next vertex can be any of those corresponding to the negative entries in the $\bar{u}$ vector. However, for all known pivot rules, there exist classes of quantifier-free $\Sigma_{\mathbb{Q}}$ -formulae on which the simplex method requires an exponential number of steps in the size of the formulae.

8.5 Summary

This chapter covers linear programming and the simplex method for solving linear programs. It covers:

- How decision procedures that reason only about conjunctive formulae extend to arbitrary Boolean structure by converting to DNF. Exercise 8.1 explores a more effective way of converting to DNF.
- A review of linear algebra.
- *Linear programs*, which are optimization problems with linear constraints and linear objective functions. Application to $T_{\mathbb{Q}}$ -satisfiability.
- The *simplex method*. Finding an initial point via a new linear program. Greedy search along vertices.

The structure of the simplex method is markedly different from the structure of the quantifier elimination procedures of Chapter 7. It focuses on the structure of the set of interpretations of the given formula, rather than on the formula itself. Exercises 8.3, 8.4, and 8.5 explore these sets, which describe polyhedra.

In addition to applications in all of engineering, arithmetical reasoning is necessary for analyzing program correctness. Of course, arithmetic usually appears in the context of other data structures in software. Chapter 10 discusses a method for combining the arithmetic decision procedures of this and the previous chapter with other decision procedures.

Bibliographic Remarks

For a presentation of linear algebra, see [42]; for a comprehensive presentation of linear and integer programming, see [82].

Work on linear programming proceeded in parallel in the western hemisphere and the USSR. Dantzig is the inventor of the simplex method in the western hemisphere [22]. Earlier work by Kantorovich in the USSR also proposes the method [44]. Weakly polynomial-time algorithms were achieved by Khachian [49] in the USSR and Karmarkar [45] in the western hemisphere.

Exercises

8.1 (★Conjunctive quantifier-free formulae). Converting an arbitrary quantifier-free Σ -formula to DNF and then applying a decision procedure to each disjunct can be prohibitively expensive. In practice, SAT solvers (decision procedures for propositional logic, such as DPLL) are used to extend a decision procedure for conjunctive quantifier-free Σ -formula to arbitrary quantifier-free Σ -formula.

- (a) Show that the DNF of a formula F can be exponentially larger than F .
- (b) Describe a procedure that, using a SAT solver, extracts conjunctive Σ -formulae from a quantifier-free Σ -formula F . Using this procedure, each discovered conjunctive formula's T -satisfiability will be decided. If it is T -satisfiable, then F is T -satisfiable, so the procedure finishes; otherwise, the procedure finds another conjunctive formula.
- (c) The proposed procedure is really no more efficient than simply converting to DNF. This part explores an optimization. An **unsatisfiable core** of T -unsatisfiable conjunctive Σ -formula G is the conjunction H of a subset of literals of G such (1) H is also T -unsatisfiable, and (2) the conjunction of each strict subset of literals of H is T -satisfiable. Improve your procedure from the previous part to use a function $\text{UnsatCore}_T(G)$ that returns a T -unsatisfiable core of a T -unsatisfiable conjunctive Σ -formula G .
- (d) Given a decision procedure DP_T for conjunctive quantifier-free Σ -formula, describe a procedure $\text{UnsatCore}(G)$ for computing an unsatisfiable core of G that takes no more than a number of DP_T calls linear in the number of literals of G . Note that G can have multiple unsatisfiable cores; your procedure need only return one.

- (e) Given a decision procedure DP_T for conjunctive quantifier-free Σ -formula, describe a procedure $\text{UnsatCore}(G)$ for computing an unsatisfiable core of G that takes no more than $O(m \log n)$ calls of DP_T , where m is the number of literals of the returned unsatisfiable core and n is the number of literals of G .

8.2 (DP for $T_{\mathbb{Q}}$). Apply the simplex method to decide the $T_{\mathbb{Q}}$ -satisfiability of the following $\Sigma_{\mathbb{Q}}$ -formulae.

- (a) $x \geq 1 \wedge 2x \leq 1$
 (b) $x + 2y \geq 1 \wedge 2x + y \geq 1 \wedge 2x + 2y \leq 1$
 (c) $x + 2y \geq 1 \wedge 2x + y \geq 1 \wedge x + y \leq 1$

8.3 (★Polyhedra & dual representation). $A\bar{x} \leq \bar{b}$ defines a subspace of $\mathbb{R}^n$ called a **polyhedron**; $A\bar{x} \leq \bar{b}$ is the **constraint representation** of this space. A polyhedron can also be described by the union of a set of vertices $V \subseteq \mathbb{R}^n$ and a set of **rays** $R \subseteq \mathbb{R}^n$; this representation is the **vertex representation**. While vertices $\bar{v} \in V$ describe points in $\mathbb{R}^n$, rays $\bar{r} \in R$ describe directions in $\mathbb{R}^n$. For example, as a vertex, $[1 \ 1]^T \in \mathbb{R}^2$ describes the point one unit right and one unit up from the origin on the plane. As a ray, it describes the “northeast” direction.

Represent each vertex $\bar{v} \in V$ and ray $\bar{r} \in R$ by the $(n+1)$ -vectors

$$\begin{bmatrix} \bar{x} \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \bar{r} \\ 0 \end{bmatrix},$$

respectively. Let P be the union of V and R represented in this way. Then for every inequality $A\bar{x} \leq \bar{b}$, there exists a vertex-ray set P with k elements such that

$$A\bar{x} \leq \bar{b} \quad \text{iff} \quad \exists \lambda_1, \dots, \lambda_k \geq 0. \quad \begin{bmatrix} \bar{x} \\ 1 \end{bmatrix} = \sum_{i=1}^k \lambda_i \bar{p}_i.$$

- (a) Given an element

$$\bar{p} = \begin{bmatrix} \bar{x} \\ \epsilon \end{bmatrix} \in P,$$

we can evaluate $A\bar{x} \leq \epsilon \bar{b}$. If $\bar{p}$ is a vertex, it is equivalent to evaluating $A\bar{x} \leq \bar{b}$, as expected. If $\bar{p}$ is a ray, it is equivalent to evaluating $A\bar{x} \leq \bar{0}$. Explain what the ray case means.

- (b) Define the **convex hull** as follows:

$$\text{hull}(P) = \left\{ \bar{x} : \exists \lambda_1, \dots, \lambda_k \geq 0. \quad \begin{bmatrix} \bar{x} \\ 1 \end{bmatrix} = \sum_{i=1}^k \lambda_i \bar{p}_i \right\}.$$

Show that $\text{hull}(P)$ is convex. Conclude that if $A\bar{x} \leq \epsilon \bar{b}$ for each $[\bar{x}^T \ \epsilon]^T \in P$, then $\text{hull}(P) \subseteq \{\bar{x} : A\bar{x} \leq \bar{b}\}$.

8.4 (★Set operations). The constraint representations of two polyhedra S_1 and S_2 are $A_1\bar{x} \leq \bar{b}_1$ and $A_2\bar{x} \leq \bar{b}_2$, respectively, and their vertex representations are P_1 and P_2 , respectively.

- (a) Write the constraint representation of $S_1 \cap S_2$ in terms of S_1 and S_2 .
- (b) Write the vertex representation of $\text{hull}(S_1 \cup S_2)$ in terms of P_1 and P_2 .
- (c) Describe how to compute whether $S_1 \subseteq S_2$. *Hint:* Use both representations.

8.5 (★Set operations with vertex representation). The vertex representations of two polyhedra S_1 and S_2 are P_1 and P_2 , respectively.

- (a) Describe how to compute whether $S_1 \subseteq S_2$ using only their vertex representations. *Hint:* Use the simplex method.
- (b) Given $A\bar{x} \leq \bar{b}$, a **vertex enumeration algorithm** lists the corresponding vertex representation. Using a vertex enumeration algorithm, describe how to compute the vertex representation of $S_1 \cap S_2$ using only their vertex representations.